

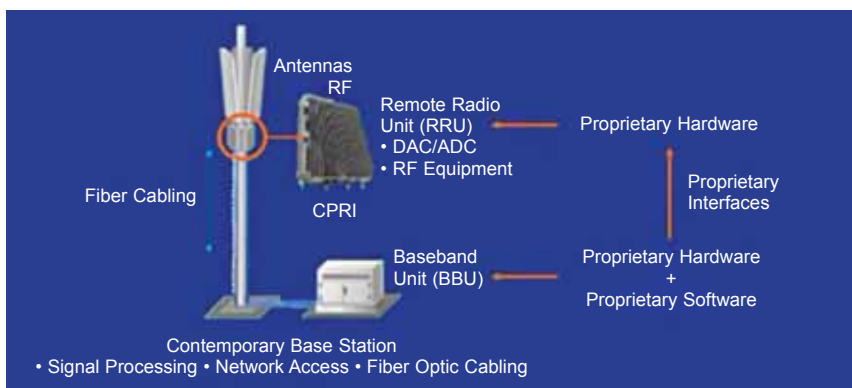


Fig. 3: Traditional RAN

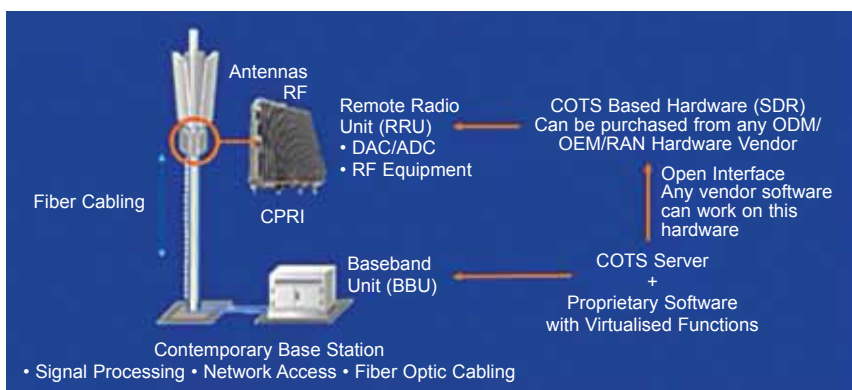


Fig. 4: Open RAN architecture

tion among the vendors themselves. Newer companies are now coming up with their own components which tend to work a lot better.

Architecture of a typical mobile network

As shown in Fig. 2, the core network and the access network are connected by the transport network, which is also known as the back-haul network. Then there is an air interface that connects a user with the service provider's network. There are various other services that run on top of the core network. The core network and the access network contain both the hardware and software that can be seen at the left in Fig. 2.

For proper understanding of the architecture let us understand a few things from previous-generation network architectures first.

All the electronics were in one or more cabinets at the bottom of a

mast (tower) when 2G and 3G were in use. The RF cables would run to the antennas on top of the mast. The RF signal continuation in the RF cable was a big problem because of the congestion that used to occur.

As a result, the later phases of 3G entering 4G allowed the industry to take a different approach that worked better. Instead of sending RF signals from the cabinet to the antennas, the RF signals were moved closer to the antennas (as can be seen in Fig. 3). This ensured that there was a very minimal loss of signals. This new piece of electronics is known as remote radio unit (RRU) or remote radio head (RRH). At the bottom we had the base band unit.

Let us move to the cabinet next. A cabinet consists of both the hardware as well as the software. RRU is always a proprietary hardware whereas BBU is a proprietary software along with the interfaces between BBU and RRU.

O-RAN has following components:

Baseband unit (BBU) in telecom systems that transports a baseband frequency, usually from a remote radio unit, to which it may be tied through optical fibre.

Remote radio unit (RRU), which is the distributed and integrated frequency unit that connects to an operator's network with the user equipment like cell phone and mobile devices.

Radio frequency (RF) band, a frequency band that is used for communication transmissions and broadcasting.

Common public radio interface (CPRI), a popular standard for transporting baseband signals to radio units. It is an interface between radio equipment control (REC) and radio equipment (RE).

Understanding the O-RAN architecture

With the O-RAN approach, the RRU is now commercial off-the-shelf (COTS) based hardware that can be purchased from any original design manufacturer (ODM) or original equipment manufacturer (OEM) or RAN hardware vendor. The BBU is proprietary software with virtualised functions running on a COTS server.

It is important to note that the interface between BBU and RRU is an open interface, which is an open functionality that we can see here (see Fig. 4). This would allow any vendor's software to work with any other vendor's hardware and vice versa.

Opportunities and challenges of O-RAN

There are quite a few opportunities offered by O-RAN, some of which are:

Avoid vendor lock-in. With the fully interoperable and desegregated RAN, the service provider should be able to choose from the best in class RAN component manufacturers.